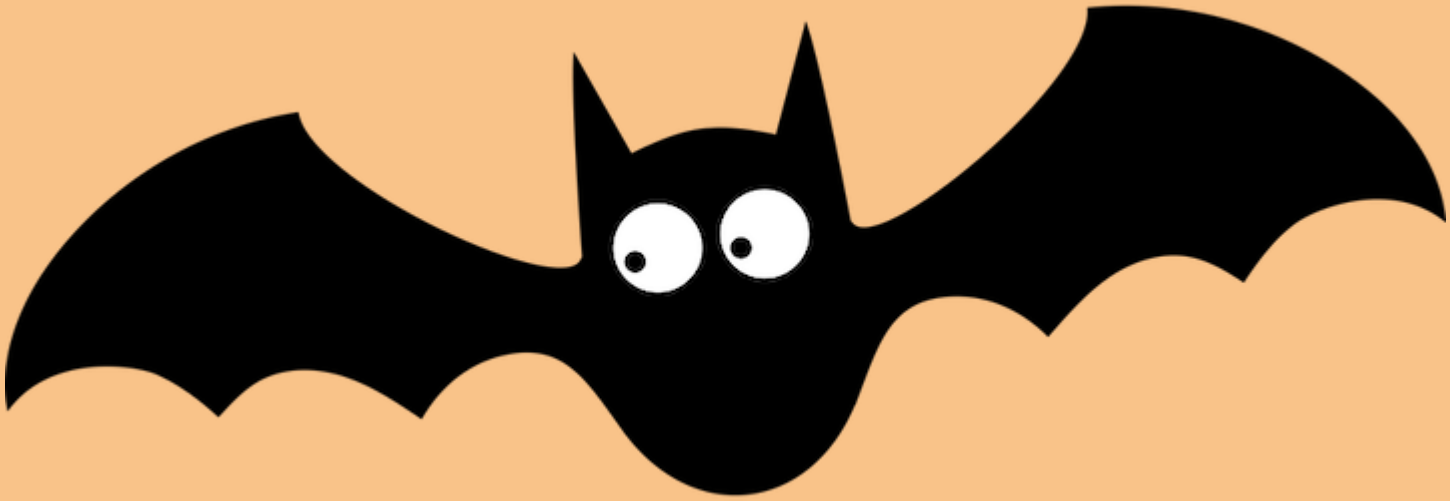


Gradient Boosting

Son Nguyen

(*)

Boosting means "learning from the mistake".

(*)

Boosting means "learning from the mistake".

- AdaBoost
- Gradient Boosting

(*)

Boosting means "learning from mistakes"

Adaboost

Increase the weights of the mistakes

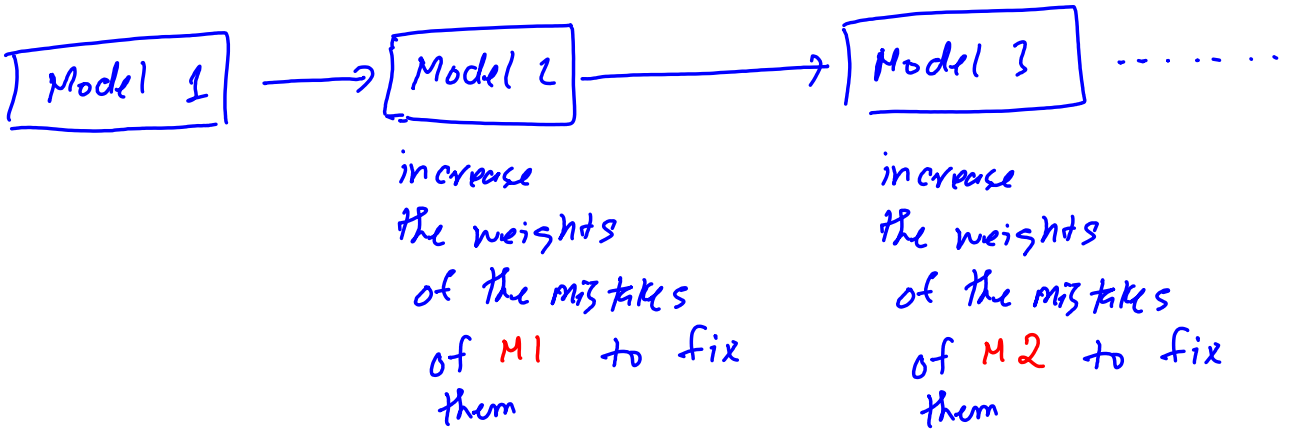
Gradient boosting

replace the original target by the erros

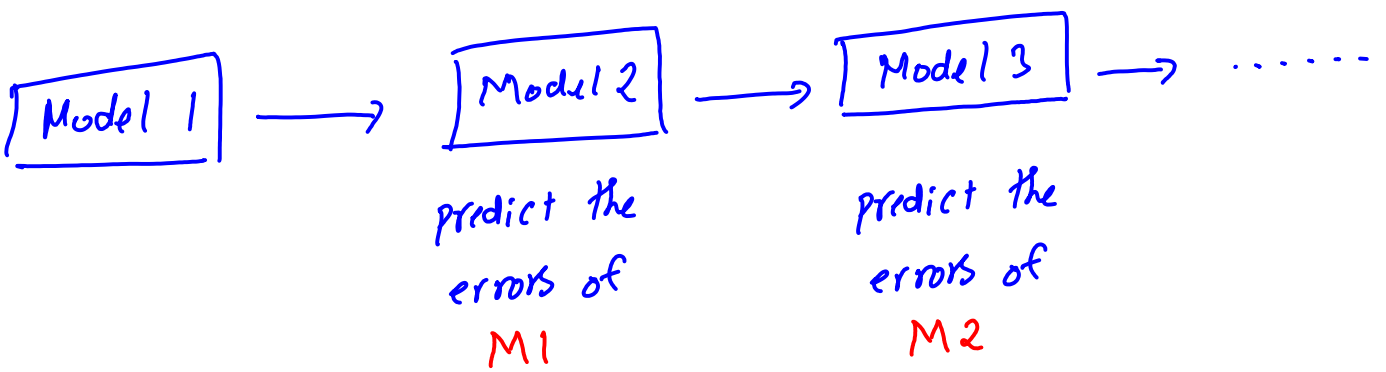
(*) Let's predict the temperature tomorrow

Model 1: predict 61°F. This model will make some error (even when error = 0).

Adaboost :



Gradient boosting



we bring model 2 just to predict the error of model 1

Model 2 will make another error. We bring in model 3
to predict the error of model 2.

And so on.

In the end the final prediction is

Predict of Model 1 + Predict of model 2 + Predict of M3

+ .....

Let's see an example:

Data :

X	Y
60	70
70	75
55	50
67	76
66	59

Data to train model 1 (M1)

l : Learning rate

X	Y	M1 predicts	error 1
60	70	40	30
70	75	90	-15
55	50	80	-30
67	76	70	6
66	59	100	41

X	error 1
60	30 . l
70	-15 . l
55	-30 . l
67	6 . l
66	41 . l

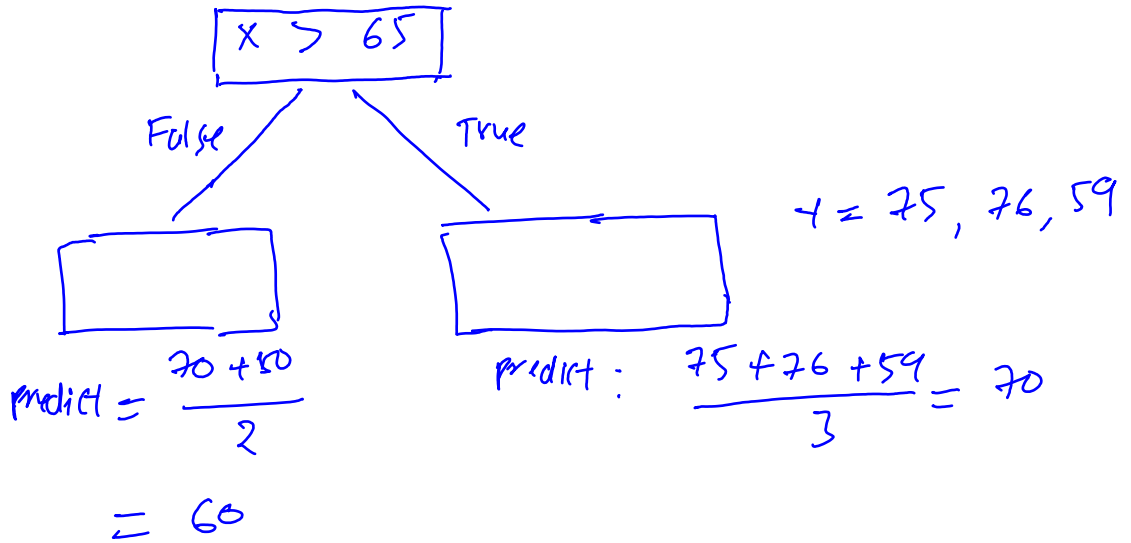
Data to train model 2

Notice that: Although Y is continuous in this case, Gradient boosting algorithm works when Y is categorical. The same goes for AdaBoost.

(*) How decision tree used for continuous Y variable.

Example:

X	Y
60	70
70	75
55	50
67	76
66	59



(*) How to evaluate the model when y is continuous

regression problem.

(1) R^2

(2) Absolute mean error (AME)

(3) MSE : Mean Squared error.

X	y	$\hat{y}$	Absolute error	Squared error
60	70	40	30	900
70	75	90	15	225

SSE = Sum squared error = 900 + 225 = 1125

MSE = Mean ——— = $\frac{1125}{2}$ = 562.5

AME = (30 + 15)/2 = 22.5

The base line model → classification: Tossing a coin and make prediction. (Random Guess)

→ Regression: $\bar{y}$ (The average of the target in training data)

$$R^2 = 1 - \frac{\text{SSE of the model}}{\text{SSE of the baseline model}}$$

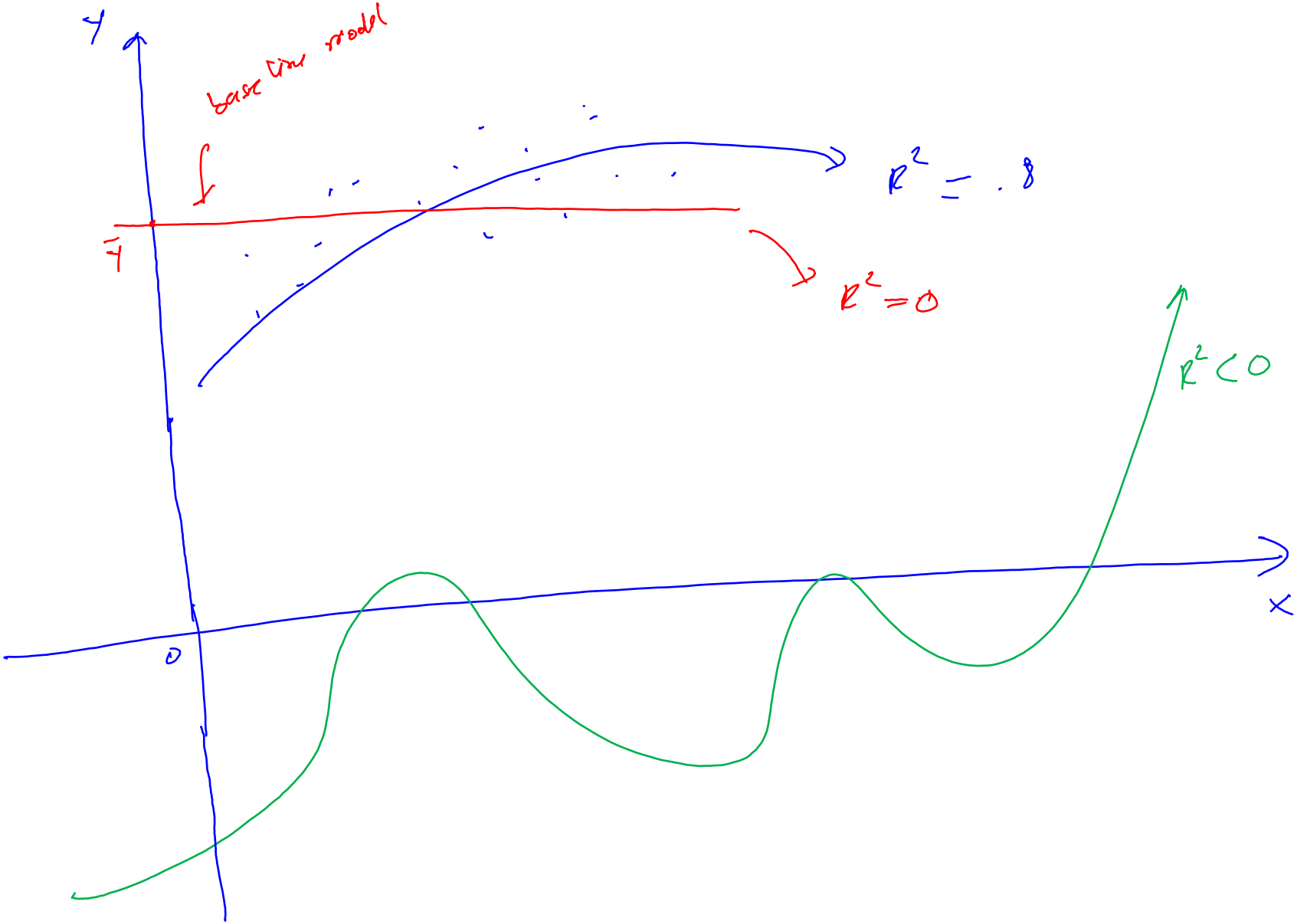
(SSE = Sum Squared Errors)

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

$R^2 = 1$: perfect model with no errors

$R^2 = 0$: when the model is baseline model. ($\bar{y}$)

$R^2 < 0$: when the model is worse than $\bar{y}$


y
baseline model
$\bar{y}$
$R^2 = .8$
$R^2 = 0$
$R^2 < 0$
x
0

(*) Hyper parameters for Adaboost and Gradient Boosting

(1) Number of weak learners

(2) Learning Rate

weak learners are usually stump or very shallow trees.